



10783 - Dynamical Mass Measurements of ngEHT/BHEX SMBH Targets with JWST/NIRSpec

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	NGC 1218	NIRSpec IFU Spectroscopy	(1) NGC-1218
	2	NGC 2663	NIRSpec IFU Spectroscopy	(2) NGC-2663
	3	NGC 3894	NIRSpec IFU Spectroscopy	(3) NGC-3894

ABSTRACT

Supermassive black holes (SMBHs) lie at the heart of galaxy evolution, yet their most massive end remains poorly constrained. This program will deliver the first dynamical SMBH mass measurements for three nearby massive radio galaxies—NGC 2663, NGC 3894, and NGC 1218—which are prime candidates for imaging by the next-generation Event Horizon Telescope (ngEHT) and the proposed Black Hole Explorer (BHEX) mission.

JWST Proposal 10783 (Created: Friday, March 13, 2026, 3:01:18PM Eastern Standard Time) - Overview

These systems host bright, compact cores and exhibit SMBH spheres of influence that are easily resolvable with JWST/NIRSpec IFU, but currently lack reliable dynamical mass determinations.

Using JWST's NIRSpec (<5 hrs total science time), we will map stellar kinematics at unprecedented precision within each black hole's sphere of influence. Combining these with archival HST imaging and well-tested pipelines (e.g., the Jeans Anisotropic Models, or JAM), we will robustly determine SMBH masses with <10% uncertainty. This will reduce the ~ 0.5 dex scatter in shadow-based mass predictions, refine angular size estimates for horizon-scale imaging, and anchor the high-mass tail of SMBH–galaxy scaling relations.

The resulting dynamical calibrations will enable cross-validation between shadow and stellar-dynamical mass measurements, guiding target selection and theoretical interpretation for upcoming ngEHT and BHEX observations. This work lays the essential groundwork for the next decade of horizon-scale black hole astrophysics, connecting galactic dynamics to the event horizon.

OBSERVING DESCRIPTION

We will obtain integral-field spectroscopy with JWST/NIRSpec to measure spatially resolved stellar kinematics within and around the black hole spheres of influence (SOI) in NGC 2663, NGC 3894, and NGC 1218. Each target will be observed using the NIRSpec IFU with the G235H/F170LP grating, which provides $R = 2700$ spectral resolution (or instrument error of approximately 47 km/s) over 1.66–3.17 micron and a $3 \text{ arcsec} \times 3 \text{ arcsec}$ field of view sampled at 0.1 arcsec per spaxel. This configuration cleanly captures the CO bandhead absorption features (2.19–2.37 micron) essential for stellar kinematic measurements, while avoiding atmospheric contamination and maximizing JWST's diffraction-limited performance.

For each galaxy, we will use the NRSIRS2 readout mode and a 4-POINT-DITHER pattern to mitigate detector artifacts, cosmic rays, and bad pixels while optimizing spatial sampling. Each observation will consist of 15–25 groups per integration and one integration per exposure. This setup results in a total exposure time tailored to each target, ranging between 1.2 and 2.0 hours, designed to achieve $S/N > 40$ per spectral pixel within the SOI. Outside the SOI, spectra will be power-binned to achieve $S/N = 30$, sufficient for mapping the larger-scale velocity field that constrains the stellar mass profile and dark matter contribution.

These NIRSpec data will yield high-fidelity 2D maps of stellar velocity and velocity dispersion, enabling Jeans Anisotropic Modeling (JAM) to measure SMBH masses with < 0.1 dex precision. The stellar light distribution for each galaxy will be modeled from archival HST imaging via the Multi-Gaussian Expansion (MGE) method, ensuring optimal dynamical modeling inputs.

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Together, these observations will deliver the first dynamical SMBH mass measurements for three of the most promising ngEHT/BHEX targets, establishing robust mass anchors at the extreme high-mass end of the BH-velocity dispersion relation and providing essential groundwork for future horizon-scale imaging of black holes.

Proposal 10783 - Targets - Dynamical Mass Measurements of ngEHT/BHEX SMBH Targets with JWST/NIRSpec

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	NGC-1218	RA: 03 08 26.2238 (47.1092658d) Dec: +04 06 39.30 (4.11092d) Equinox: J2000	Proper Motion RA: -0.495 mas/yr Proper Motion Dec: -0.26000004709203495 mas/yr Parallax: 3.472E-4" Epoch of Position: 2000	
	<i>Comments: This object was generated by the target selector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> <i>Category=Galaxy</i> <i>Description=[Active galactic nuclei, Lenticular galaxies, Quasars, Radio galaxies, Seyfert galaxies]</i> <i>Extended=YES</i>				
	(2)	NGC-2663	RA: 08 45 8.1444 (131.2839350d) Dec: -33 47 41.06 (-33.79474d) Equinox: J2000	Proper Motion RA: 0.3 mas/yr Proper Motion Dec: 0.1 mas/yr Epoch of Position: 2000	
	<i>Comments: This object was generated by the target selector and retrieved from the SIMBAD database.</i> <i>Category=Galaxy</i> <i>Description=[Active galactic nuclei, Elliptical galaxies, Quasars, Radio galaxies]</i> <i>Extended=YES</i>				
	(3)	NGC-3894	RA: 11 48 50.3582 (177.2098258d) Dec: +59 24 56.38 (59.41566d) Equinox: J2000	Proper Motion RA: 0.2 mas/yr Proper Motion Dec: -0.1 mas/yr Epoch of Position: 2000	
	<i>Comments: This object was generated by the target selector and retrieved from the SIMBAD database.</i> <i>Category=Galaxy</i> <i>Description=[Active galactic nuclei, Elliptical galaxies, LINER galaxies, Radio galaxies, Seyfert galaxies]</i> <i>Extended=YES</i>				

Proposal 10783 - Observation 1 - Dynamical Mass Measurements of ngEHT/BHEX SMBH Targets with JWST/NIRSpec

Observation	Proposal 10783, Observation 1: NGC 1218										Fri Mar 13 20:01:18 GMT 2026	
	Diagnostic Status: Warning											
Diagnostics	Observing Template: NIRSpec IFU Spectroscopy											
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	NGC-1218	RA: 03 08 26.2238 (47.1092658d) Dec: +04 06 39.30 (4.11092d) Equinox: J2000			Proper Motion RA: -0.495 mas/yr Proper Motion Dec: -0.26000004709203495 mas/yr Parallax: 3.472E-4" Epoch of Position: 2000						
	Comments: This object was generated by the target selector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Galaxy Description=[Active galactic nuclei, Lenticular galaxies, Quasars, Radio galaxies, Seyfert galaxies] Extended=YES											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	1 NGC-1218	WATA	FULL	F110W	NRSRAPID	3	1	1	42.947	274382	
Template	HFF Readout Mode											
	false											
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235H/F170LP	NRSIRS2	15	1	false	true	NONE	4	4	4435.023	274382

Proposal 10783 - Observation 2 - Dynamical Mass Measurements of ngEHT/BHEX SMBH Targets with JWST/NIRSpec

Observation	Proposal 10783, Observation 2: NGC 2663											Fri Mar 13 20:01:18 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(2)	NGC-2663	RA: 08 45 8.1444 (131.2839350d)				Proper Motion RA: 0.3 mas/yr					
			Dec: -33 47 41.06 (-33.79474d)				Proper Motion Dec: 0.1 mas/yr					
			Equinox: J2000				Epoch of Position: 2000					
	Comments: This object was generated by the target selector and retrieved from the SIMBAD database. Category=Galaxy Description=[Active galactic nuclei, Elliptical galaxies, Quasars, Radio galaxies] Extended=YES											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	2 NGC-2663	WATA	FULL	F110W	NRSRAPID	3	1	1	42.947	274382	
Template	HFF Readout Mode											
	false											
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235H/F170LP	NRSIRS2	20	1	false	true	NONE	4	4	5893.912	274382

Proposal 10783 - Observation 3 - Dynamical Mass Measurements of ngEHT/BHEX SMBH Targets with JWST/NIRSpec

Observation	Proposal 10783, Observation 3: NGC 3894											Fri Mar 13 20:01:18 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(3)	NGC-3894	RA: 11 48 50.3582 (177.2098258d)				Proper Motion RA: 0.2 mas/yr					
			Dec: +59 24 56.38 (59.41566d)				Proper Motion Dec: -0.1 mas/yr					
			Equinox: J2000				Epoch of Position: 2000					
	Comments: This object was generated by the target selector and retrieved from the SIMBAD database. Category=Galaxy Description=[Active galactic nuclei, Elliptical galaxies, LINER galaxies, Radio galaxies, Seyfert galaxies] Extended=YES											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	3 NGC-3894	WATA	FULL	F110W	NRSRAPID	3	1	1	42.947	274382	
Template	HFF Readout Mode											
	false											
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235H/F170LP	NRSIRS2	25	1	false	true	NONE	4	4	7352.801	274382